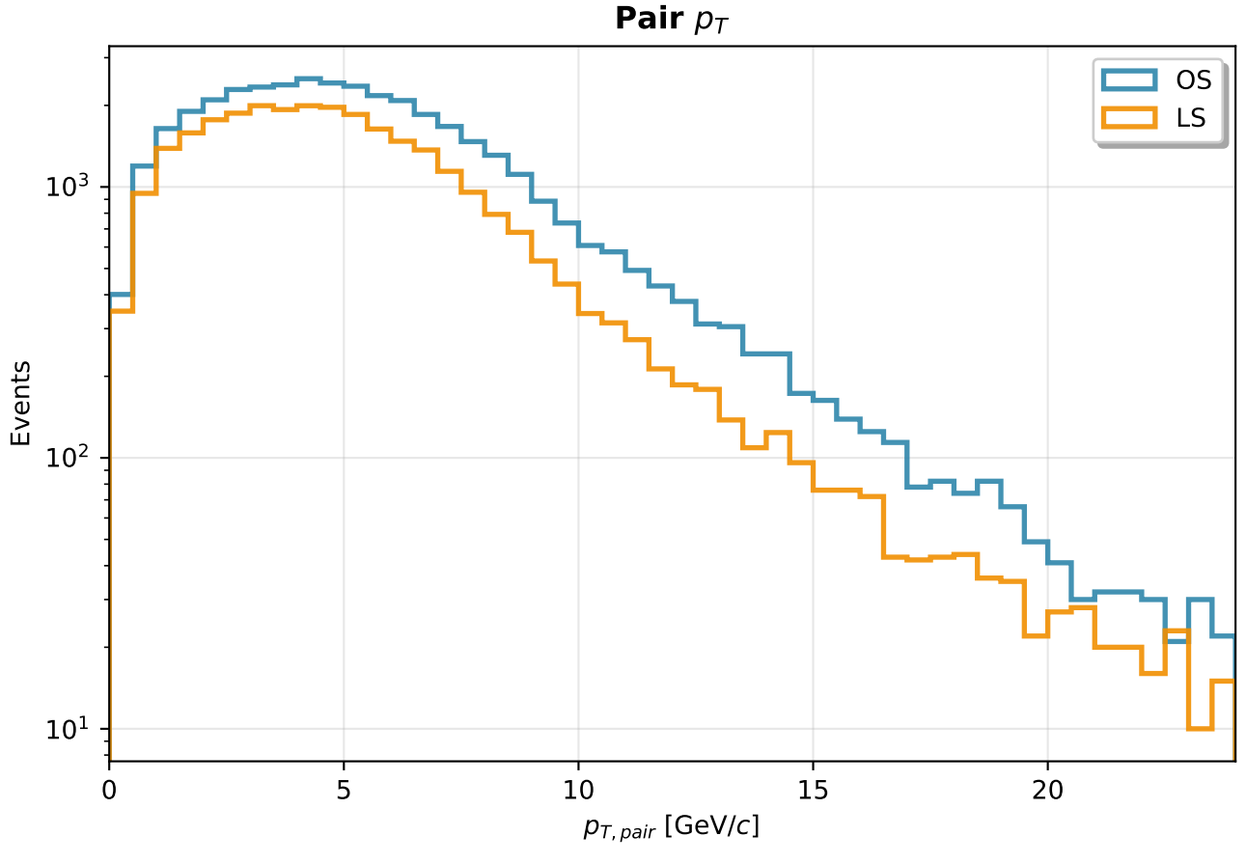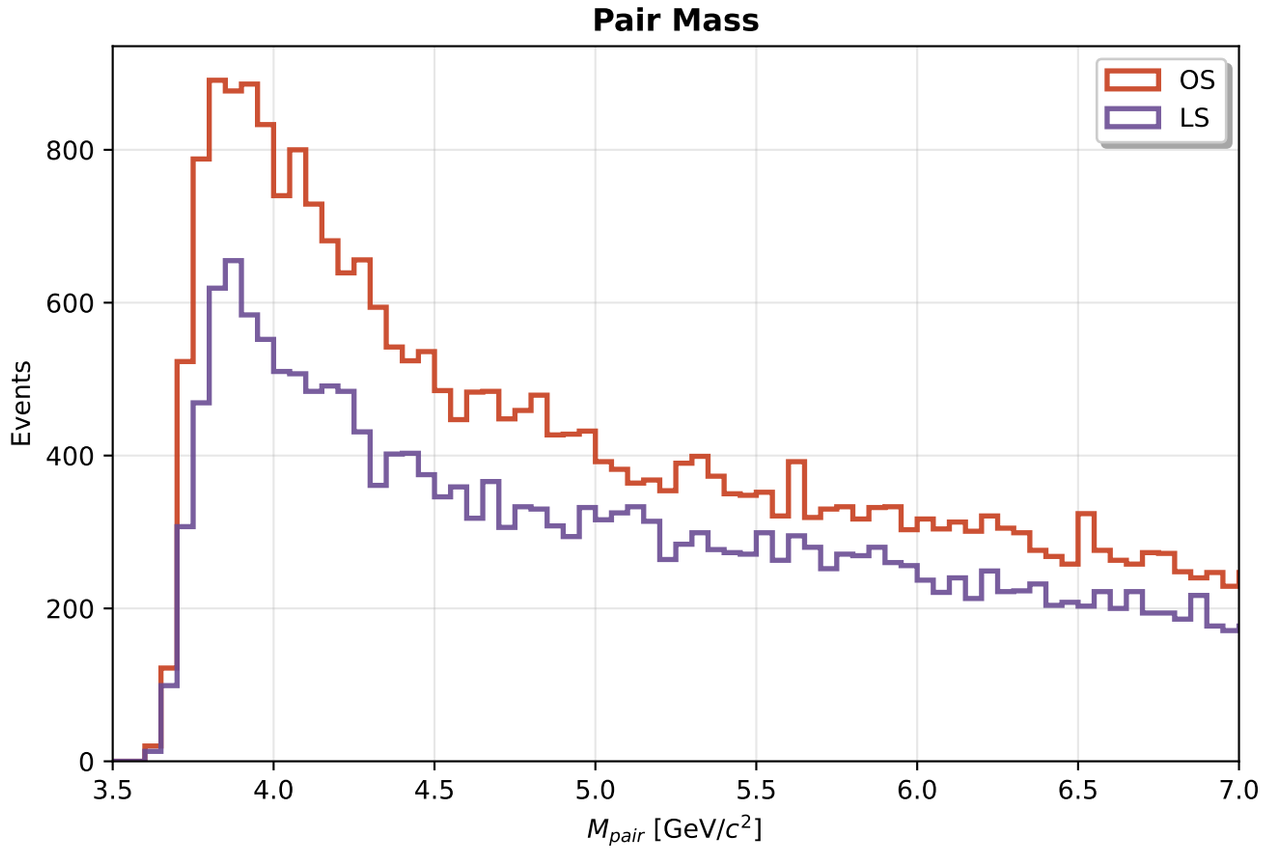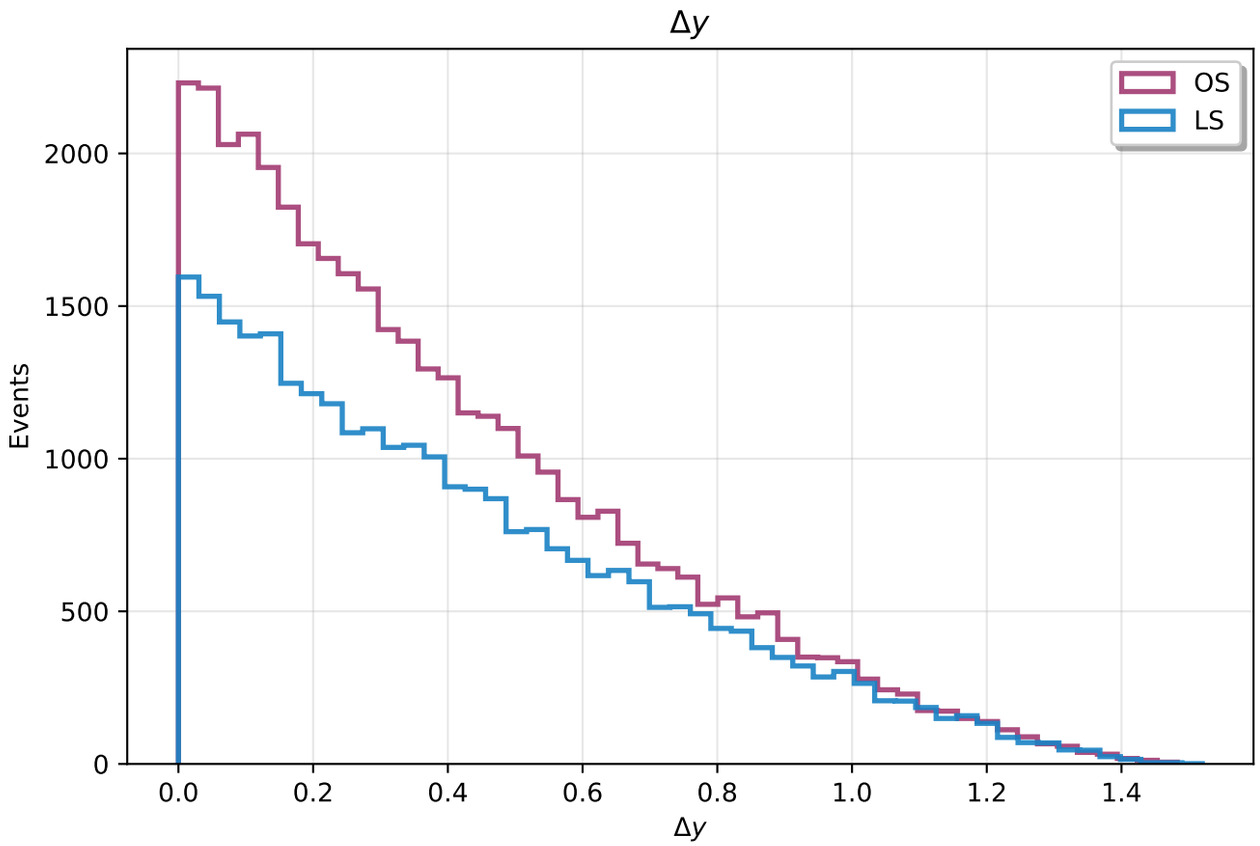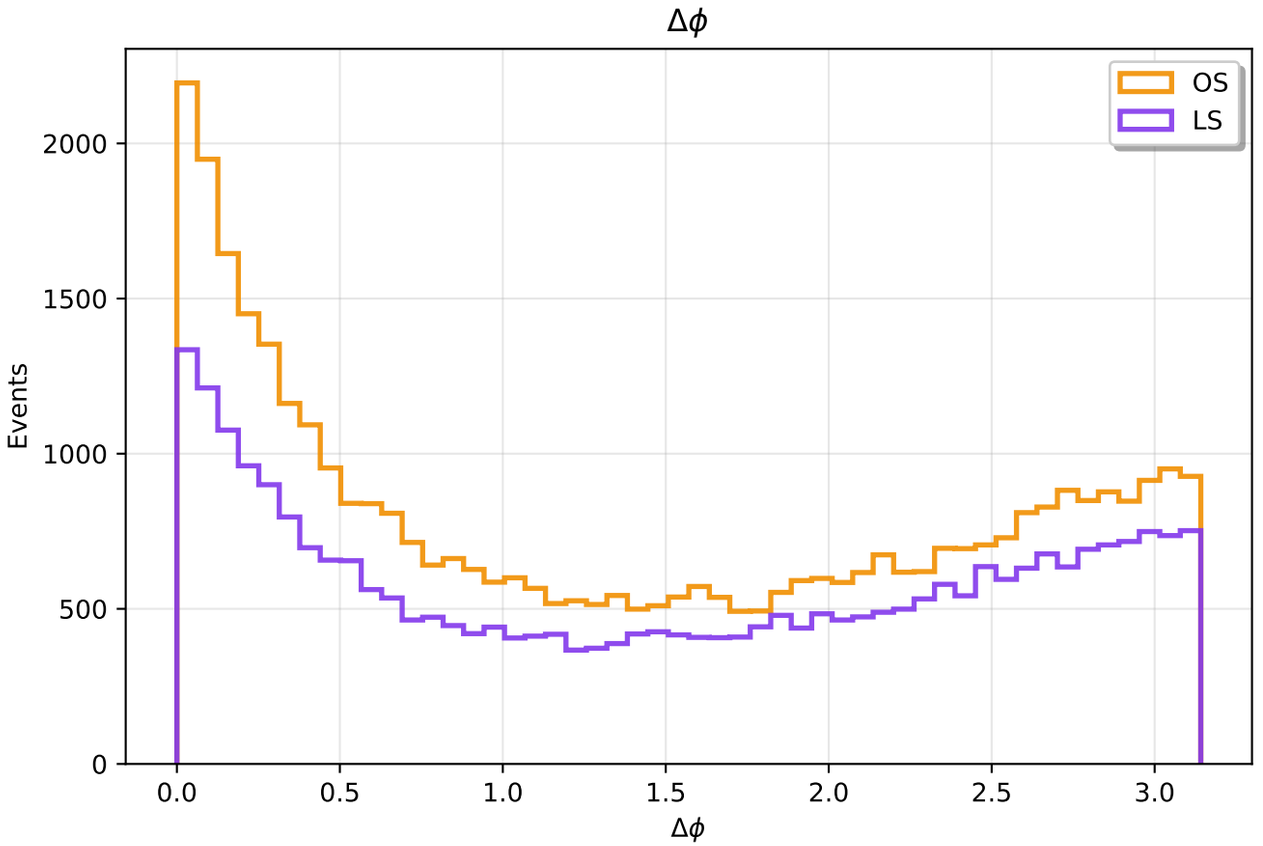OS vs LS Pair Kinematics Comparison in the 3σ region